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Name	: Ms. SATVIKI PATIDAR	**UHID**	: MGMNM9000701818
Age/Gender	: 7 Y 10 M 10 D / Female		
Consultant	: Dr. Roopesh Kumar V R		
Passport No	: 0	**Encounter No**	: IP 902604240000
Lab No	: 922771000004		
Sample Collected	: 29/04/2026 21:34:09		
Sample Received	: 29/04/2026 21:36:26		
Sample Reported	: 12/05/2026 16:52:32	**Ward**	: B Type 10th South/1002
Immunohisto No	: H2601270 & IHC26-141		

IMMUNOHISTOCHEMISTRY

Neuro Test Package (HPE with IHC Neuro Panel)

SPECIMEN:

LEFT THALAMIC LESION.

CLINICAL DETAILS:

Ms. Satviki Patidar, a 6-year-old female child, known case of a left thalamopeduncular/optic pathway hypothalamic glioma with mesial temporal extension (modified DODGE type 3B, H+), previously underwent left pterional craniotomy with microsurgical excision of the mesial temporal component on 22/04/2025. Histopathological evaluation from multiple institutional reviews demonstrated a diffusely infiltrating pediatric-type low-grade glial neoplasm with astrocytic morphology, characterized by moderate cellularity, mild-to-moderate nuclear pleomorphism, and absence of definite high-grade features such as necrosis, significant mitotic activity, or microvascular proliferation. Immunohistochemistry showed positivity for GFAP and OLIG2, retained ATRX expression, retained H3K27me3 expression with H3K27M negativity, IDH1-R132H negativity, p16 retention without evidence of CDKN2A/B homozygous deletion, TP53 wildtype expression pattern, low proliferative index (MIB-1 approximately 3-5%), and MGMT promoter non-methylated status. Molecular profiling revealed a pathogenic DBN1::BRAF fusion, consistent with MAPK pathway alteration, supporting the integrated diagnosis of a pediatric-type diffuse low-grade glioma, MAPK pathway-altered (diffuse infiltrating astrocytic glioma). DNA methylation profiling was advised for further molecular classification refinement. The

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patient was subsequently initiated on Tovorafenib therapy in March 2026. Following the second dose, she developed progressive neurological deterioration with right-sided hemiparesis, restricted extraocular movements, dysphagia, drowsiness, and reduced responsiveness. MRI brain with DTI revealed interval increase in tumor size with obstructive hydrocephalus and raised intracranial pressure, necessitating elective intubation on 24/04/2026. CT brain confirmed obstructive hydrocephalus, following which she underwent right Keen's point programmable ventriculo-peritoneal shunt placement on 27/04/2026 with interval reduction in hydrocephalus. Repeat MRI brain (navigation protocol) dated 29/04/2026 demonstrated interval enlargement of the left hemispheric lesion measuring approximately 5.9 × 5.9 × 5 cm with central necrotic change, patchy hemorrhagic foci, and reduced hydrocephalus status post shunting. Following multidisciplinary discussion, the patient underwent left pterional and high frontal craniotomy with neuronavigation and intraoperative neurophysiological monitoring-guided safe maximal resection of the left thalamic lesion on 29/04/2026.

GROSS DESCRIPTION:

Received in a formalin-filled container labeled with the patient's name (SATVIKI PATIDAR), age/sex (7 YEARS/FEMALE), and UHID (9000701818), designated as "LEFT THALAMIC LESION," are multiple irregular grey-tan soft tissue fragments collectively measuring approximately 4 cc in aggregate.
Entire specimen submitted for histopathological examination.

SECTION KEY:

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A-B: ENTIRE SPECIMEN.

Grossed by: DR. KSS / SN / PHP

MICROSCOPIC DESCRIPTION:

A,B. Sections from the left thalamic lesion demonstrate fragments composed predominantly of neoplastic tissue with adjacent cortical parenchyma showing patchy involvement by an infiltrating glial neoplasm, likely centered within the subcortical white matter with extension into the overlying cortex. The tumor exhibits a diffusely infiltrative growth pattern with focal perineuronal and perivascular satellitosis. Overall tumor cellularity is variable, ranging from moderately cellular areas to focal hypercellular nodules composed of spindle to polygonal neoplastic cells exhibiting moderate nuclear pleomorphism, hyperchromasia, and occasional conspicuous nucleoli. In the moderately cellular regions, tumor cells are arranged in diffuse sheets within a fibrillary background and display predominantly astroglial morphology, characterized by variably enlarged ovoid to elongated spindled nuclei with coarse chromatin. Patchy areas demonstrate bipolar piloid morphology with focal gemistocytic differentiation. The vascular architecture is predominantly composed of delicate branching capillaries; however, focal areas show subtle early microvascular proliferation. Scattered foci of hemorrhage and patchy microcalcifications are identified. Focal areas suspicious for incipient necrosis are also noted, although well-formed pseudopalisading necrosis is not identified. Mitotic activity is focal and reaches up to 2 mitoses per 10 high-power fields in the most proliferative regions. No Rosenthal fibers, eosinophilic granular bodies, myxoid microcystic change, dysplastic neurons, or neurocytic/ependymal rosettes are identified. Focal perivascular lymphocytic and neutrophilic inflammatory infiltrates are present.

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Overall, the histomorphological features are those of a diffusely infiltrating glioma with astroglial differentiation, exhibiting variable cellularity, focal proliferative activity, and early incipient high-grade features.

IMMUNOHISTOCHEMISTRY PANEL FINDINGS

- **GFAP:** POSITIVE IN TUMOR CELLS, SUPPORTING GLIAL DIFFERENTIATION.
- **OLIG2:** POSITIVE, CONSISTENT WITH GLIAL LINEAGE.
- **IDH1 (R132H):** NEGATIVE; NO EVIDENCE OF CANONICAL IDH1 R132H MUTATION, FAVORING IDH-WILDTYPE STATUS.
- **ATRX:** RETAINED NUCLEAR EXPRESSION, ARGUING AGAINST IDH-MUTANT ASTROCYTOMA.
- **NEUROFILAMENT (NF):** HIGHLIGHTS PRESERVED AXONAL PROCESSES AND DELICATE FIBRILLARY NEUROPIL BETWEEN TUMOR CELLS, SUPPORTING AN INFILTRATIVE GROWTH PATTERN.
- **H3K27ME3:** RETAINED NUCLEAR EXPRESSION IN TUMOR CELLS, SUGGESTING PRESERVED PRC2 FUNCTION.
- **H3K27M:** NEGATIVE; NO EVIDENCE OF H3K27M MUTATION, ARGUING AGAINST DIFFUSE MIDLINE GLIOMA, H3 K27-ALTERED.
- **P53:** PREDOMINANTLY DIFFUSE MODERATE NUCLEAR POSITIVITY IDENTIFIED IN APPROXIMATELY 51-60% OF TUMOR CELLS.
- **BRAF V600E:** NEGATIVE IN TUMOR CELLS.

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- **CD34:** NEGATIVE IN TUMOR CELLS.
- **KI-67 (MIB-1) PROLIFERATION INDEX:** APPROXIMATELY 6–9% IN HOT-SPOT AREAS.

FINAL IMPRESSION:

LEFT THALAMIC LESION, LEFT PTERIONAL HIGH FRONTAL CRANIOTOMY WITH SAFE MAXIMAL RESECTION:

PEDIATRIC-TYPE DIFFUSE INFILTRATING GLIOMA WITH ASTROGLIAL DIFFERENTIATION, IDH-WILDTYPE AND H3K27-ALTERED NEGATIVE, SHOWING FOCAL EARLY HIGH-GRADE/ANAPLASTIC FEATURES INCLUDING SUBTLE MICROVASCULAR PROLIFERATION, FOCI SUSPICIOUS FOR INCIPIENT NECROSIS, INCREASED P53 EXPRESSION, AND ELEVATED KI-67 LABELING INDEX (6–9% IN HOT-SPOT AREAS) (KNOWN CASE OF DBN1::BRAF FUSION-POSITIVE, MAPK PATHWAY-ALTERED PEDIATRIC-TYPE DIFFUSE LOW-GRADE GLIOMA, CNS WHO 5TH EDITION).

EXPANDED NEXT-GENERATION SEQUENCING (NGS) AND GENOME-WIDE DNA METHYLATION PROFILING ARE RECOMMENDED FOR DEFINITIVE INTEGRATED CLASSIFICATION, MOLECULAR RISK STRATIFICATION, AND ASSESSMENT FOR POSSIBLE THERAPY-ASSOCIATED MOLECULAR EVOLUTION.

COMMENT:

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The current resection specimen demonstrates a diffusely infiltrating pediatric-type glioma with astroglial differentiation showing interval histomorphological progression compared to the previously reported low-grade glial neoplasm, including increased cellularity, focal hypercellular nodules, subtle early microvascular proliferation, foci suspicious for incipient necrosis, increased p53 expression, and elevated proliferative activity (Ki-67 labeling index approximately 6–9% in hotspot areas). **In the present post-Tovorafenib therapy setting, these findings are concerning and raise the possibility of evolving tumor progression and/or therapy-associated biological changes.** The rapid clinical and radiological progression shortly after initiation of Tovorafenib therapy, coupled with the emergence of focal early microvascular proliferation and increased proliferative activity, suggests either an accelerated biological course or a heterogeneous therapeutic response within the tumor. Although the overall features remain within the spectrum of a MAPK pathway-altered pediatric-type diffuse glioma in the background of a previously characterized DBN1::BRAF fusion-positive neoplasm, **the emergence of focal early high-grade features warrants comprehensive molecular re-evaluation.**

It is noteworthy that, while the category of pediatric-type diffuse low-grade gliomas predominantly comprises CNS WHO grade 1 neoplasms, **the biological behavior and assigned grade frequently correlate with the underlying molecular driver alteration.** Rearrangement/fusion-driven tumors are generally enriched for lower-grade histology and more indolent clinical behavior, whereas tumors driven by activating single nucleotide variants may demonstrate comparatively higher proliferative potential and a greater likelihood of WHO grade 2 morphology. Importantly, contemporary evidence suggests that molecular risk stratification may serve as a more powerful predictor of outcome than histological grade alone.

Furthermore, given that DBN1::BRAF represents a relatively uncommon fusion partner compared to the more

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typical KIAA1549::BRAF fusion, additional molecular re-evaluation for co-occurring alterations, including CDKN2A/B loss, PI3K/AKT/mTOR pathway alterations, or other resistance-associated genomic events, is strongly recommended, as such alterations may contribute to resistance to RAF-monotherapy and facilitate biological progression or anaplastic transformation.

In this clinical context, expanded Next-Generation Sequencing (NGS) and genome-wide DNA methylation profiling are strongly recommended for definitive integrated classification. These studies play critical and complementary roles in confirming tumor lineage, identifying additional oncogenic drivers, resistance-associated molecular alterations, or epigenetic evolution, and refining prognostic stratification. Integrated molecular and methylation-based profiling may further help classify the tumor into low-, intermediate-, or high-risk biological categories, which represent major determinants of clinical outcome and are essential for guiding subsequent therapeutic decision-making, including continuation or modification of targeted therapy and consideration of additional adjuvant treatment strategies.

IMMUNOHISTOCHEMISTRY (IHC) - METHODOLOGY:

Immunohistochemical studies were performed on formalin-fixed, paraffin-embedded tissue sections using validated antibody clones on an automated immunostaining platform (Roche Ventana Benchmark GX), following standardized laboratory protocols and manufacturer-recommended procedures.

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ANTIBODY PANEL UTILIZED:

- **ATRX** – Clone D-5
- **BRAF V600E** – Clone VE1
- **CD34** – Clone QBEND10
- **GFAP** – Clone EP13
- **OLIG2** – Clone EP112
- **IDH1 (R132H)** – Clone H09
- **P53** – Clone BP53-11
- **KI-67 (MIB-1)** – Clone 30-9
- **NEUN** – Clone A60
- **H3K27M** – Clone RM192
- **H3K27ME3** – Clone MD48R
- **NEUROFILAMENT** – Clone 2F11

A polymer-based detection system was employed for antigen visualization. Appropriate internal and external positive and negative controls were included with each staining run and demonstrated expected immunoreactivity, thereby validating assay specificity, sensitivity, reproducibility, and staining adequacy.

*** End Of Report ***

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IMMUNOHISTOCHEMISTRY

Authorized By
Dr Sunil Kumar K S, MD
PATHOLOGIST
Senior Consultant

Verified By
MGMPF655/Dr Sunil Kumar

Disclaimer:
Slides and blocks will be stored for 5 years. Surgical specimens will be stored for 1 month from the date of the report.
Blocks and slides shall be provided only upon an official written request.